

Exercise 1: Control Structures

```
CREATE TABLE CUSTOMERS (  
    customer_id NUMBER PRIMARY KEY,  
    age NUMBER,  
    loan_interest_rate NUMBER,  
    balance NUMBER,  
    is_vip VARCHAR2(10)  
);
```

```
CREATE TABLE LOANS (  
    loan_id NUMBER PRIMARY KEY,  
    customer_id NUMBER,  
    due_date DATE  
);
```

```
INSERT INTO CUSTOMERS VALUES (1, 65, 0.05, 12000, 'FALSE');  
INSERT INTO CUSTOMERS VALUES (2, 45, 0.04, 5000, 'FALSE');  
INSERT INTO LOANS VALUES (101, 1, SYSDATE + 10);  
COMMIT;
```

Scenario 1

```
BEGIN  
    FOR cust IN (SELECT * FROM CUSTOMERS) LOOP  
        IF cust.age > 60 THEN  
            UPDATE CUSTOMERS  
                SET loan_interest_rate = loan_interest_rate - 0.01  
                WHERE customer_id = cust.customer_id;  
        END IF;  
    END LOOP;  
    COMMIT;  
END;  
/
```

Scenario 2

```
BEGIN  
    FOR cust IN (SELECT * FROM CUSTOMERS) LOOP  
        IF cust.balance > 10000 THEN  
            UPDATE CUSTOMERS  
                SET is_vip = 'TRUE'  
                WHERE customer_id = cust.customer_id;  
        END IF;  
    END LOOP;  
    COMMIT;  
END;
```

/

Scenario 3

```
SET SERVEROUTPUT ON;
BEGIN
    FOR ln IN (SELECT * FROM LOANS WHERE due_date BETWEEN
SYSDATE AND SYSDATE + 30) LOOP
        DBMS_OUTPUT.PUT_LINE('Reminder: Loan for customer ' ||
ln.customer_id || ' is due soon.');
```

END LOOP;

```
END;
/
```

Exercise 3: Stored Procedures

Scenario 1

```
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest IS
BEGIN
    UPDATE CUSTOMERS
    SET balance = balance * 1.01;
    COMMIT;
END;
/
```

Scenario 2

```
CREATE TABLE EMPLOYEES (
    employee_id NUMBER PRIMARY KEY,
    department_id NUMBER,
    salary NUMBER
);
```

```
INSERT INTO EMPLOYEES VALUES (101, 10, 5000);
COMMIT;
```

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(p_dept_id IN
NUMBER, p_bonus_pct IN NUMBER) IS
BEGIN
    UPDATE EMPLOYEES
    SET salary = salary + (salary * p_bonus_pct / 100)
    WHERE department_id = p_dept_id;
    COMMIT;
END;
/
```

Scenario 3

```
CREATE OR REPLACE PROCEDURE TransferFunds(  
    p_from_acc IN NUMBER,  
    p_to_acc IN NUMBER,  
    p_amount IN NUMBER  
) IS  
    v_balance NUMBER;  
BEGIN  
  
    SELECT balance INTO v_balance FROM CUSTOMERS WHERE  
customer_id = p_from_acc;  
  
    IF v_balance >= p_amount THEN  
        UPDATE CUSTOMERS SET balance = balance - p_amount WHERE  
customer_id = p_from_acc;  
        UPDATE CUSTOMERS SET balance = balance + p_amount WHERE  
customer_id = p_to_acc;  
        COMMIT;  
    ELSE  
        DBMS_OUTPUT.PUT_LINE('Insufficient funds for  
transfer.');
```

transfer.');

```
    END IF;  
END;  
/
```